

Which Class Did Better?

AP Statistics · Unit 1 · Topic 1.9 · TEACHER KEY

CED TETHER

- **Skill 2.D** — Compare distributions or relative positions of points within a distribution.
- **Enduring Understanding UNC-1** — Graphical representations and statistics allow us to identify and represent key features of data.
- **LO UNC-1.N** — Compare graphical representations for multiple sets of quantitative data. [Skill 2.D]
- **EK UNC-1.N.1** — Any of the graphical representations, e.g., histograms, side-by-side boxplots, etc., can be used to compare two or more independent samples on center, variability, clusters, gaps, outliers, and other features.
- **LO UNC-1.O** — Compare summary statistics for multiple sets of quantitative data. [Skill 2.D]
- **EK UNC-1.O.1** — Any of the numerical summaries (e.g., mean, standard deviation, relative frequency, etc.) can be used to compare two or more independent samples.

Essential question: What does ‘did better’ mean for a whole class, and which feature of a plot answers it?

DOK-3 skill: 4.B · **FRQ pattern:** compare-two-boxplots-adjudicate-competing-claims-then-choose-one-statistic

DOK rationale: Part (c) requires weighing two competing claims about which class did better against several features of two boxplots at once, then committing to a single statistic and defending it; either claim can be supported by one feature alone, so the answer depends on coordinating center, variability, and an outlier rather than on any single procedure.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- How many students does the top of a whisker represent? How many does the median line represent?
- If the 40 in Period B were a 33 instead, would Sam’s claim change? Would Tia’s?
- Point at the box widths. Which class would you rather be a random student in? Why?
- Your parent asks ‘how did the class do?’ — which number answers that question?

[WARN] LOOK FOR

- Sentences that describe one plot at a time with no comparison word.
- Reading the whisker end as ‘most students’.
- Choosing the mean for Period B without noticing the outlier at 5.

SCORING PART (c) — E / P / I

Expected elements

- **tia-center** (required) — Chooses Tia using medians 31 versus 27

- **compares-spread** (required) — Uses IQR 8 versus 11, overall ranges 20 versus 35, or B's low outlier as further evidence
- **median-over-max** (required) — Chooses median and explains a maximum describes an extreme observation rather than the class center

E	All three elements are correct and linked to this problem. Accept equivalent plain-language reasoning.
P	At least one element includes a correct evidence-to-reason link, but another is missing or incorrect.
I	No correct evidence-to-reason link; only a choice, copied numbers, or unsupported statements.

Common mistakes

- Reading the maximum as ‘the class did better’
- Describing each boxplot separately instead of comparing (no comparative language)
- Calling the wider box ‘more scores’ rather than ‘more variable’
- Choosing the mean for Period B without noticing the outlier at 5 pulls it down

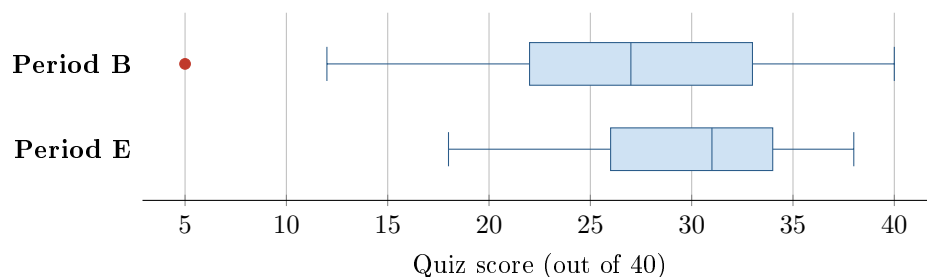
Optional #1 — not collected

DOK 2 Two bus routes were timed for 20 mornings each. Route 1: mean 24.0 min, SD 6.5 min, median 22 min. Route 2: mean 24.5 min, SD 1.8 min, median 24 min. Write two sentences comparing the routes for a student who must not be late, using comparative language and the statistics given.

Key: The means are nearly the same (24.0 versus 24.5 minutes), while Route 2 has much lower SD (1.8 versus 6.5 minutes), so it is more consistent. A student concerned about unpredictable travel could prefer Route 2, but these summaries alone do not give the worst delay or the probability of being late.

[STAR] THE PROBLEM — annotated

Two AP Statistics classes took the same 40-point quiz. The side-by-side boxplots show the scores. Two students read them differently. **Sam:** “Period B did better — its highest score is 40 and Period E’s is only 38.” **Tia:** “Period E did better — its median is higher and its scores are more consistent.”



FIRST TAKE — one sentence before you work the parts

Which class did better on the quiz? One sentence, and name ONE feature of the plot you used.

Key: Not graded. Expect a split toward Sam (the 40 is eye-catching); that split is the lesson.

[BOOK] COMPARING DISTRIBUTIONS (keep on the page)

Use comparative words such as higher, lower, more variable, or similar. Compare center (median), variability ($IQR = Q_3 - Q_1$, $range = \text{maximum} - \text{minimum}$), apparent shape, and unusual features in context. Include plotted outliers when finding the overall range. A boxplot hides values within each quartile, so qualify shape claims with ‘appears.’ A median describes center; a maximum describes the highest observed value.

DOK 1 (a) Read the median for each class.

Key: Period B: 27 points. Period E: 31 points.

DOK 2 (b) In two or three sentences, compare the quiz scores’ center, spread, apparent shape, and unusual features. Use points and comparative words.

Key: E has a higher median (31 versus 27 points) and a smaller IQR (8 versus 11 points). B’s overall range is $40 - 5 = 35$ points, versus E’s $38 - 18 = 20$, and B has the low outlier at 5. B is fairly balanced within its box but extends farther on the low side overall; E also appears more spread toward low scores. Boxplots cannot establish detailed shape.

DOK 3 (c) Whose claim is better supported, Sam’s or Tia’s? Cite two specific features of the boxplots that settle it. Then: if you could report only ONE statistic per class to a parent asking ‘how did the class do?’, which statistic would you choose, and why that one rather than the maximum?

Key: Tia’s. Two settling features: the medians (31 vs. 27 — the typical student in E scored higher) and the spread (E’s IQR 8 vs. B’s 11, plus B’s outlier at 5). Sam’s evidence is a single value: one student in B scored 40, which says nothing about the class. Report the median: it describes the typical student and is less affected by extreme scores; the maximum describes exactly one student and would make Period B look better because of one paper.